Yousef Alaa Awad

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Education

University of Central Florida — Orlando, FL
Bachelor of Science in Computer Engineering with Honors | GPA: 3.77 — Aug 2027
Relevant Courses: Verification of Digital Systems, HDL in Digital Systems Design, & Adv. Computer Architecture
Affiliations: IEEE-HKN (Zeta Chi Chapter), IEEE, ACM

Work Experience

Lockheed Martin Corporation — Orlando, FL
ASIC/FPGA Design CWEP Intern — Feb 2026 – July 2026

- Developed a Formal Property Verification (FPV) flow in Synopsys VC Formal, authoring 20+ SVA for bounded model checking; captured 3 critical bugs (illegal state transitions, deadlocks) and validated with 80%+ coverage.
- Extended FPV flow with Formal Security Verification (FSV), authoring security properties for a complex module to formally verify against unauthorized state access and security violations.
- Deployed a reproducible FPV/FSV sign-off suite in GitLab CI/CD with parallel matrix jobs, architecting a modular workflow for new-module integration and standardizing verification across 3 ASIC teams.

Embedded Software Engineering CWEP Intern — Apr 2025 – Feb 2026

- Designed a system-level Hardware-in-the-Loop (HIL) simulation environment in C++, enabling high-fidelity testing into already existing workflow for 20+ software engineers by implementing a CIGI v2 compliant image generator.
- Remediated 250+ security violations flagged by static code analysis, resolving critical memory-safety issues including buffer overflows and unsafe pointer access across the codebase.
- Engineered the migration of legacy C++/Qt tooling environments from Visual Studio to a CMake build system, slashing compilation times by 50% and establishing a modular architecture to improve future scalability of the application.

Projects

KnightCore – RISC-V SoC – 2025 AMD Hardware Competition – *Group Project* — Apr 2025 – Aug 2025

- Developed a coverage-driven verification testbench in Python (Cocotb) to validate a custom 32-bit RISC-V SoC (RV32I ISA), implementing constrained-random stimulus to effectively explore the design state space.
- Developed a comprehensive test suite with directed tests for all ISA-defined instructions and constrained-random stimulus generation to stress the ALU and branch prediction logic, leading to the discovery and resolution of 5 critical RTL bugs.

2025 SouthEastCon Robotics Competition – Embedded Software Lead – *Group Project* — Sep 2024 – Apr 2025

- Led software pipeline for 10+ team using ROS2, enabling low-latency control between modular hardware and decision nodes. Used Gazebo and OpenCV for validation, contributing to 1st place in Design and 2nd in Performance at competition.
- Developed embedded firmware on Teensy 4.1 for sensor fusion and actuator control with modular, real-time C++ code. Optimized for high-frequency loops and low resource use, improving system responsiveness and performance.

CyndaQuil Compiler/Language – *Solo Project* — Sep 2024 – Jan 2025

- Designed and implemented a statically-typed compiled language from scratch in C++, utilizing CMake to manage the build system across parsing, semantic analysis, and x86 assembly generation stages.
- Engineered low-level system components and optimized performance, integrating Assembly and C to enhance hardware-software interfacing, ensuring minimal latency and efficient computation for embedded systems.

Leadership

Eta Kappa Nu (IEEE-HKN)
Chapter President, Zeta Chi Chapter — Apr 2026 – Present

- Directing overall chapter operations and strategic vision, overseeing the executive board to ensure the continued growth and national recognition of the Zeta Chi Chapter.

Chapter Vice President, Zeta Chi Chapter — Apr 2025 – Apr 2026

- Lead the re-establishment of the dormant IEEE-HKN chapter, positioning the organization to achieve the national "*Key Chapter*" award by growing membership to 31 students and developing a new recruitment pipeline.

Institute of Electrical and Electronics Engineers
Treasurer, UCF Student Branch — Apr 2025 – Apr 2026

- Managed a $10,000+ chapter budget and facilitated over $6,500 in sponsorship funding, directly supporting student project development and participation in competitions like SouthEastCon.

Technical Skills

Formal Verification: Synopsys VC Formal, SystemVerilog Assertions (SVA), Bounded Model Checking, Formal Property Verification (FPV), Formal Security Verification (FSV)
Languages: SystemVerilog, Verilog, C++, Tcl, Python, C, Bash, Rust, Assembly (x86, RISC-V, MIPS)
Computer Architecture: RISC-V ISA, AXI/Memory-Mapped Interfaces, Pipelining, Caching, Memory Hierarchy
Hardware & EDA Tools: Synopsys VC Formal, Synopsys VCS, Xilinx Vivado, Verilator
Systems & Scripting: Linux, Git, GitLab CI/CD, ROS/ROS2, Real-Time C++, CMake/Make